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Definition of the Hypothalamic GnRH Pulse Generator in mice

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Abstract

The pulsatile release of luteinizing hormone (LH) is critical for mammalian fertility. However, despite several decades of investigation, the identity of the neuronal network generating pulsatile reproductive hormone secretion remains unproven. We use here a variety of optogenetic approaches in freely-behaving mice to evaluate the role of the arcuate nucleus kisspeptin (ARN^{KISS}) neurons in LH pulse generation. Using GCaMP6 fibre photometry, we find that the ARN^{KISS} neuron population exhibits brief (~1 min) synchronized episodes of calcium activity occurring as frequent as every 9 min in gonadectomized mice. These ARN^{KISS} population events were found to be near-perfectly correlated with pulsatile LH secretion. The selective optogenetic activation of ARN^{KISS} neurons for 1 min generated pulses of LH in freely-behaving mice, whereas inhibition with archaerhodopsin for 30 min suppressed LH pulsatility. Experiments aimed at re-setting the activity of the ARN^{KISS} neuron population with halorhodopsin were found to re-set on-going LH pulsatility. These observations indicate the ARN^{KISS} neurons as the long-elusive hypothalamic pulse generator driving fertility.

Significance statement

Neural networks located in the hypothalamus are responsible for generating ultradian patterns of hormone secretion that control a wide variety of functions. How these neural networks generate pulsatile hormone secretion remain unknown. We report here that a population of hypothalamic kisspeptin neurons represent the gonadotropin-releasing hormone (GnRH) pulse generator. These cells have the remarkable ability to generate synchronized GnRH secretion every 9 minutes to drive pulsatile gonadotropin hormone secretion in the blood. These observations indicate the arcuate kisspeptin neurons as the origin of reproductive hormone pulsatility in mice and offer the prospect of better understanding and manipulating fertility in the clinic.

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Endocrine signaling from the pituitary gland occurs primarily as pulsatile hormone secretion. For most hormones, this pulsatility is thought to be generated by episode generators located within the hypothalamus and pituitary gland (1, 2). However, with few exceptions, it has been extremely difficult to determine the identity and nature of these neuroendocrine pulse generators (3, 4). Although the critical importance of pulsatile hormone secretion for reproduction was discovered in the 1970s (5, 6), the identity of the so-called “gonadotropin-releasing hormone (GnRH) pulse generator” (7, 8) responsible for pulsatile gonadotropin secretion remains unknown.

The discovery of the critical importance of kisspeptin signalling within the brain of humans and rodents for fertility (9-11) has suggested a potential role for kisspeptin signaling in the generation of pulsatile hormone secretion. Humans and rodents with global deletions or mutations in kisspeptin or the kisspeptin receptor fail to exhibit normal pulsatile luteinizing hormone (LH) secretion (9, 10, 12, 13). Of the two kisspeptin neuron populations that innervate GnRH neurons, particular attention has focussed upon the arcuate nucleus kisspeptin (ARN^{KISS}) neurons in terms of pulse generation (14, 15). Investigators have shown that ARN^{KISS} neurons co-express a variety of different neurotransmitters including glutamate, neurokinin B and dynorphin (15, 16), for which these cells also express functional receptors (17, 18). Alongside anatomical evidence, it has been suggested that the ARN^{KISS} neurons may exhibit synchronized behaviour that could intermittently activate GnRH neurons and, thus, act as a pulse generator (15, 16, 19). This attractive hypothesis remains unproven.

We employ here a range of optogenetic approaches in freely-behaving mice to evaluate whether the ARN^{KISS} neurons represent the GnRH pulse generator. Using GCaMP fiber photometry, we observed brief repetitive episodes of elevated calcium within the ARN^{KISS} neuron population and find a near-perfect correlation of these events with pulsatile LH secretion. Mimicking these brief synchronization events through selective channelrhodopsin (ChR2)-activation of ARN^{KISS} neurons was sufficient to generate pulsatile LH secretion while optogenetic inhibition of ARN^{KISS} neurons suppressed pulsatile LH secretion. Further, re-setting the activity of the ARN^{KISS} neuron population was found to re-set pulsatile LH secretion. Together these observations indicate that the ARN^{KISS} neurons are the GnRH pulse generator.

RESULTS

ARN^{KISS} neurons exhibit episodic activity in vivo

We used fiber photometry (20) to examine whether ARN^{KISS} neurons exhibited intermittent synchronous activity in awake freely behaving mice. We targeted the calcium reporter GCaMP6s to ARN^{KISS} neurons by bilateral injection of an AAV9-CAG-Flex-GCaMP6s into the ARN of male KISS1-Cre;tdTomato reporter mice (21).

Immunohistochemical analyses showed that GCaMP6 was expressed in kisspeptin neurons throughout the ARN with 51-67% of ARN^{KISS} neurons expressing GCaMP6 depending on rostro-caudal level (Suppl Fig.1). Dual-labeled cells represented 89±6% to 96±6% of all GCaMP6-expressing cells in the ARN (Suppl Fig.1). Acute brain slice experiments using cell-attached recording combined with GCaMP6 calcium imaging showed that GCaMP6 faithfully reports on the electrical activity of ARN^{KISS} neurons in these mice (6 cells from 4 mice; Fig.1B).

We next injected a group of KISS1-Cre male mice with GCaMP6 AAV and implanted a 400µm-diameter optical fiber just above the ARN (Fig.1A). Mice were gonadectomised (GDX) to generate a state in which the GnRH pulse generator is operating at its highest frequency. After three weeks of daily handling and habituation, the mice were connected to the patch cord of a fibre photometry system and calcium signals measured for up to 4h in freely-behaving mice. Abrupt increases in calcium signal were detected at approximately regular intervals of 8.9±0.5 min (range; 4.5-20.7 min) in all mice examined (n=7; Fig.1C). Histological analysis showed that all optical fibers were located above the middle to caudal aspect of the ARN. Photometry sampling revealed a consistent temporal dynamic for calcium events with an initial rapid increase over 19.6±2.4s followed by a slower decrease that, together, lasted for 74.2±6.6s (16 events from 7 mice; Fig.1D,E).

To demonstrate that the GCaMP6 signal reflects the activity of ARN^{KISS} neurons, we co-injected Cre-dependent AAVs for both GCaMP6s and hM3Dq (22) into the ARN of GDX KISS1-Cre mice. Three to 4 weeks later, calcium imaging experiments in acute brain slices demonstrated that addition of 1-10µM clozapine N-oxide (CNO) to the bathing medium resulted in a marked and prolonged increase in both firing and intracellular calcium concentrations ([Ca²⁺]_i) in all ARN^{KISS} neurons (4 cells dual recorded, 38 cells in total, 3 mice; Fig.1F). KISS1-Cre mice injected with GCaMP6s and hM3Dq were then prepared with a photometry fibre directed at the ARN. Peripheral administration of CNO (i.p., 2mg/kg) to freely-behaving mice was found to rapidly (within 5 min) enhance basal [Ca²⁺]_i by 33±8% and increase the frequency of calcium events by 100±52% (p=0.04, paired t-test; n=4; Fig.1G).

These studies indicate that ARN^{KISS} neurons exhibit episodes of synchronized activity every ~9 min. This was surprising as previous estimates of GnRH pulse generator activity in GDX mice, obtained from measuring plasma luteinizing hormone (LH) levels (23), indicated a GnRH/LH pulse interval of >20 min. We next examined the relationship of ARN^{KISS} neuron calcium events to pulsatile LH secretion by taking 3-min tail-tip blood samples for a 60-80 min period while also monitoring calcium signals. This revealed that calcium events were almost perfectly correlated with increments in LH secretion; 94% of calcium events were followed by a >10% increase in LH concentration. No increases in LH were detected without a preceding calcium event (60 LH pulses in 7 mice; Fig.1H & I). The interval from the peak of the calcium event to the peak of the LH pulse was highly consistent across all animals being 4.00±0.14min (n=7). A significant temporal relationship existed between calcium and LH peaks (P<0.05, repeated measures ANOVA), while no such relationship existed between randomly-selected LH samples and calcium peaks (Suppl.Fig.2).

Brief activation of ARN^{KISS} neurons generates LH pulses in vivo

We have previously demonstrated that optogenetic activation of ARN^{KISS} neurons for >2 min at 10Hz generates pulse-like increments in LH secretion in anaesthetized mice (24). However, the fibre photometry studies above clearly demonstrate that ARN^{KISS} neuron calcium events last for only ~1 min. Hence, we examined whether brief 1 min periods of ARN^{KISS} neuron activation were sufficient to evoke an LH pulse in awake freely-behaving mice.

Channelrhodopsin (ChR2) was targeted to ARN^{KISS} neurons by unilateral injection of a AAV9-EF1-dflox-hChR2-mCherry into the ARN of intact male and female KISS1-Cre mice (24). Intact mice exhibit an LH pulse every 90-180 min (12, 23) thereby providing a long window of basal LH secretion for optogenetic stimulation experiments. Activation of ARN^{KISS} neurons for 1 min (10Hz, 473nm light, 5mW) was found to reliably evoke pulse-like increments in LH secretion in conscious male (n=5; Fig.2A-C) and female (n=6; Fig.2D,E) mice and this was repeatable at 30 min intervals. Compatible with sex differences in endogenous LH pulse amplitude (12, 23), optogenetic activation of ARN^{KISS} neurons evoked 0.5-1ng amplitude LH pulses in females (Fig.2D,E) and ~4ng/ml amplitude pulses in males (Fig.2B,C). The profile of LH secretion evoked by 10Hz 1min stimulation was very similar to the profile of endogenous LH pulses observed in intact male mice (Fig.2F).

The inhibition of ARN^{KISS} neuron firing suppresses pulsatile LH secretion in vivo

To examine the necessity of ARN^{KISS} neurons for pulsatile LH secretion we employed an optogenetic archaerhodopsin (ArchT)-mediated silencing approach (25, 26). Bilateral injections of a Cre-dependent ArchT-tdTomato AAV were made into the ARN of adult female KISS1-Cre;GFP reporter mice that were also GDX at the time of surgery to generate high frequency pulsatile LH secretion. This resulted in 60-69% of ARN^{KISS} expressing ArchT (Fig.3B & Suppl. Fig.3), representing 77-83% of all ArchT-expressing neurons depending on rostro-caudal location within the ARN (Suppl Fig.3). The use of a GFP reporter for labelling kisspeptin neurons is not as efficient as the tdTomato reporter (GCaMP6 and eNPHR3.0 experiments) and, consequently, under-reports the number of ARN^{KISS} neurons (Suppl. Figs 1,3,4).

Cell-attached recordings from ARN^{KISS} neurons in acute brain slices prepared from ArchT AAV-injected KISS1-Cre mice demonstrated that 532nm green light promptly suppressed firing rates by 76±8% over the 5-min period with a gradual return to baseline firing when illumination was stopped (11 cells from 5 mice; Fig.3C,D). As ARN^{KISS} neurons exhibit very low levels of spontaneous electrical activity in the brain slice, 10-20nM neurokinin B was added to the bathing medium to increase their firing (27).

In awake mice (n=6), bilateral green light activation of ArchT in ARN^{KISS} neurons for a 30-min period resulted in a 49% suppression of LH pulse frequency (p<0.01;

repeated measures ANOVA with *post-hoc* Dunnett's tests), 65% decrease in the amplitude of pulses occurring during illumination ($p < 0.05$; ANOVA with *post-hoc* Dunnett's tests) and a 49% decline in basal LH secretion (Fig.3E,F,I-K). Controls ($n=9$) included blue light illumination in four mice previously showing LH pulse inhibition with green light (Fig.3G), green light illumination of wild-type mice injected with the Cre-dependent ArchT AAV (Fig.3H) and Kiss-Cre mice without ArchT. In all cases, laser illumination had no effect on LH secretion (Fig.3L-N).

Re-setting the activity of ARC^{KISS} neurons re-sets pulsatile LH secretion in vivo

The above studies show that ARC^{KISS} neuron activity is both necessary and sufficient for pulsatile LH secretion in the mouse. However, it remains possible that these cells represent a relay of the pulse generator rather than being the episode generator themselves. In the course of developing the optogenetic methodology for this study we also examined halorhodopsin (eNpHR3.0) as a potential inhibitory tool. Electrophysiological studies on eNpHR3.0-expressing ARC^{KISS} neurons revealed that 532nm green light evoked a mean $81 \pm 4\%$ inhibition, and often complete silencing, of NKB-activated ARC^{KISS} neurons but then resulted in a marked rebound activation for up to 2 min when illumination was stopped (Fig.4C,D). This represented an ideal way to temporally re-set the activity of the ARC^{KISS} neuron population *in vivo*. As such, we undertook a second series of optogenetic inhibition studies using eNpHR3.0 to both confirm the ArchT result and also determine whether we could re-set LH pulsatility through manipulation of only the ARC^{KISS} neurons.

Bilateral injections of Cre-dependent eNpHR3.0 AAV were made into the ARN of adult GDX female KISS1-Cre mice. This resulted in 63-70% of ARC^{KISS} neurons expressing eNpHR3.0 (Fig.4B, Suppl Fig.4) with dual-labelled neurons accounting for up to 94% of all eNpHR3.0 cells depending on rostro-caudal location within the ARN (Suppl. Fig.3).

Bilateral green light activation of eNpHR3.0 in ARC^{KISS} neurons resulted in the suppression of pulsatile LH secretion for the 30-min duration of illumination (Fig.4E,F; $n=8$). This resulted in a 66% decrease in LH pulse frequency ($p < 0.05$; ANOVA with *post-hoc* Dunnett's tests) with 5 of 8 mice exhibiting a complete suppression of pulsatile LH release during the 30-min illumination period (Fig.4E,F). The remaining 3 mice exhibited a single pulse during this time with, overall, a 65% reduction in LH pulse amplitude ($p < 0.01$; ANOVA with *post-hoc* Dunnett's tests; Fig.4J). Control mice ($n=7$) including wild-type mice injected with eNpHR3.0 AAV and KISS1-Cre mice with off-target fiber optic placements, did not exhibit any changes in pulsatile or basal LH secretion during green light illumination (Fig.4G,H,L-N). Four mice with unilateral activation of eNpHR3.0 in ARC^{KISS} neurons did not exhibit any changes in LH (Suppl. Fig.5E).

As predicted, a robust LH pulse was generated immediately after termination of the green light illumination in eNpHR3.0 KISS1-Cre mice (Fig.4.E,F,I), but not in controls (Fig.4G,H,L). KISS1-Cre mice also exhibited a re-setting of on-going LH pulsatility (Fig.4E,F). As observed with mean LH values (Fig.4I), the random timing of LH pulses in individual mice results in an averaged flat line before a fall

in mean LH levels upon illumination and the initiation of an LH pulse when the laser is turned off. Subsequent to this evoked pulse, another was found to occur 35 min later in all mice (Fig.4I arrow). Control mice, and indeed ArchT KISS1-Cre mice (Fig.3I), do not show any relationship between LH pulsatility and green light termination (Fig.4G,H,L). For unknown reasons, the expression of eNpHR3.0 in ARN^{KISS} neurons generates mice with large, relatively slow LH pulses independent of any illumination. As such, LH pulses occur every 34 min in these mice (Fig.4K) and this is entirely consistent with the on-going 35 min pulse interval found after re-setting LH pulse frequency with eNpHR3.0 (Fig.4I).

In undertaking the eNpHR3.0 experiments, we generated cohorts of KISS1-Cre mice in which eNpHR3.0 was expressed throughout the ARN but optical fibers located bilaterally in either the rostral (n=5) or middle/caudal (N=7) ARN (Suppl. Fig.5). Whereas light activation of the middle/caudal ARN^{KISS} neurons was always effective in suppressing pulsatile LH secretion, illumination of the rostral ARN^{KISS} had no (Suppl. Fig.5F), or only partial (Suppl. Fig.5G), effects on pulsatile LH secretion.

DISCUSSION

We demonstrate here that the ARN^{KISS} neurons are almost certainly the long-elusive hypothalamic GnRH pulse generator. These neurons exhibit episodes of synchronized activity lasting ~1min that are almost perfectly correlated with pulsatile LH secretion. Consistent with this, the activation of ARN^{KISS} neurons for 1 min reliably generated pulses of LH secretion. Using two inhibitory optogenetic strategies, we show that the selective inhibition of ARN^{KISS} neurons activity results in the suppression of pulsatile LH secretion. Furthermore, the re-setting of ARN^{KISS} neuron activity is found to re-set pulsatile LH secretion.

The calcium events detected with fiber photometry very likely represent the synchronized electrical activity of the ARN^{KISS} neurons. In brain slice experiments, we found that GCaMP6 reliably reported ARN^{KISS} neuron firing. Furthermore, co-expression of GCaMP6 and hM3Dq in ARN^{KISS} neurons demonstrated that the CNO-induced activation of firing in ARN^{KISS} neurons was associated with large increases in [Ca²⁺]_i within individual ARN^{KISS} neurons. We noted that, *in vivo*, CNO raised the basal calcium signal and doubled the frequency of synchronized calcium events. This shows that the ARN^{KISS} neuron population can maintain an episodic synchronized output in the presence of persistent Gq-protein-coupled receptor activation. This did not occur in the acute brain slice where CNO evoked a long-lasting activation of both firing and [Ca²⁺]_i in individual ARN^{KISS} neurons with no evidence of episodic activity. This possibly results from the absence of network activity in the acute brain slice where the deafferented ARN^{KISS} neurons show little spontaneous activity and no episodic or synchronous firing (27, 28).

The ARN^{KISS} neuron population exhibits robust increases in calcium approximately every 9 min. This was unexpected given that prior assessments of pulse generator activity had indicated LH pulse interval to be >20 min (29). Nevertheless, we find here that 94% of synchronized calcium events are followed by an increment in LH secretion in gonadectomized mice. Often these changes in

LH did not occur as a classical pulse but could be observed, for example, as an increment in LH on the downward shoulder of a prior LH pulse. These events are not detected by current pulse analysis algorithms and indicate the need for the further development of LH pulse analysis methodologies. In terms of ARN function, it is intriguing to find a 10-min neuronal oscillator embedded amongst ARN neural circuitries subserving longer-term function such as feeding initiation (30). With evidence for kisspeptin inputs into these feeding circuits and others (31), it will be intriguing to examine the impact of this fast oscillator on other ARN outputs in future studies.

The selective inhibition of ARN^{KISS} neuron activity with both ArchT and eNpHR3.0 optogenetic strategies suppressed pulsatile LH secretion. We found that the eNpHR3.0 approach generated a stronger inhibition of ARN^{KISS} neuron firing *in vitro* and this correlated with a more robust inhibition of LH pulsatility compared with ArchT. It is likely that a sub-population of the ~1,000 ARN^{KISS} neurons in the mouse (21) are required for pulse generation. Only ~70% of ARN^{KISS} neurons expressed ArchT or eNpHR3.0 in our experiments and fibre optic locations in the rostral ARN were ineffective in suppressing pulsatile LH secretion. This raises the possibility that a sub-population of kisspeptin neurons located in the mid-caudal ARN may comprise the GnRH pulse generator. How few ARN^{KISS} neurons may be required for pulsatility is not known but a 30% suppression of total *Kiss1* mRNA expression in the ARN does not appreciably alter LH pulsatility in rats (32). Unilateral suppression of ARN^{KISS} neuron activity was insufficient to alter LH pulsatility, whereas unilateral activation generated LH pulses. Contralateral projections between the bilateral ARN^{KISS} neuron networks may maintain their synchronization (18, 33), although a unilateral population is sufficient to generate an LH pulse.

Although the Chr2 and ArchT optogenetic studies demonstrated the necessity and sufficiency of ARN kisspeptin neurons for pulse generation, they did not establish whether this population was, itself, the pulse generator. The silencing followed by rebound activation of ARN^{KISS} after eNpHR3.0 activation provided a useful tool to explore this issue. Rebound activation has been reported following eNpHR, but not ArchT, activation in other neural populations and may result from chloride ion-based changes in GABA_A reversal potential (34). We predicted that if ARC^{KISS} neurons were simply a relay within the pulse generator, then eNpHR3.0-mediated synchronous activation would have no impact upon LH pulsatility beyond the initiated pulse. On the other hand, if the ARC^{KISS} neurons were the genesis of pulsatility this randomly-timed activation should be able to re-set future pulsatile LH secretion. This was indeed found to be the case with synchronous eNpHR3.0 activation generating an LH pulse upon offset as well as a subsequent time-locked spontaneous LH pulse ~ 35 min later in every mouse. This also occurred regardless of when the stimulus was applied in relation to endogenous LH pulsatility. We were surprised to find that the expression of eNpHR3.0 in ARN^{KISS} neurons resulted in the slowing of LH pulse frequency to ~34 min independent of any light activation. Light-independent effects of eNpHR3.0 on neuronal behaviour have not been reported previously but, nevertheless, suggest the possibility that changes in ARN^{KISS} neuron chloride ion homeostasis may be one strategy for slowing pulse generator frequency.

Tracing studies in mice have demonstrated that ARN^{KISS} neurons only project to the distal projections of GnRH neurons in the vicinity of the median eminence (35). These distal GnRH neuron projections, termed dendrons, are unusual in that they display both dendrite- and axon-like properties and are regulated directly by multiple neural inputs (36, 37). We show here that ARN^{KISS} neurons release kisspeptin for ~1min to generate an LH pulse and recent studies have shown that brief (10-60s) applications of kisspeptin to the median eminence increase intracellular calcium levels in GnRH neuron dendrons and evoke GnRH secretion independent of action potential firing in GnRH neurons (37, 38). This ARN^{KISS}→GnRH neuron dendron pathway is also likely to exist in primates as the closest associations between kisspeptin and GnRH fibers occur in the internal zone of the median eminence of the monkey (39) and kisspeptin levels within the ARN/median eminence fluctuate and can occur coincident with episodes of pulsatile GnRH secretion in this species (40). The ability of ARN^{KISS} neurons to target the merged distal projections of GnRH neurons around the median eminence abrogates the need for a pulse generator to project to the widely-dispersed GnRH neuron cell bodies. This represents an atypical mechanism for rhythm generation in the brain (41) and emphasizes the diversity of mechanisms that the neuroendocrine hypothalamo-pituitary axes likely employ to drive the pulsatile hormone secretion.

In conclusion, we provide here robust evidence for the identity of the GnRH pulse generator proposed over three decades ago. It remains that the mechanisms responsible for the synchronization of ARN^{KISS} neurons are unknown. Similarly, it is unclear how gonadal steroids and neuronal inputs proposed to mediate multiple different homeostatic and external influences upon pulsatility (14) interact with the ARN^{KISS} neurons. It is expected that the further elucidation of both the intra- and inter-cellular mechanics of the ARN^{KISS} pulse generator will provide a robust framework for advancing strategies aimed at regulating LH pulse frequency in the clinic.

METHODS

Animals:

Adult littermate heterozygous KISS1-Cre^{+/-} or KISS1-Cre^{-/-} (21) mice, alone or crossed onto Cre-dependent reporter ROSA26-CAG- τ GFP or Ai9-CAG-tdTomato lines (mixed 129S6Sv/Ev C57BL6 genetic background) were housed under a 12:12h lighting schedule (lights on at 6:00 A.M.) with *ad libitum* access to food and water. All experiments were approved by the University of Otago Animal Welfare and Ethics Committee.

Stereotaxic injections of AAVs:

Recombinant Cre-dependent AAVs encoding GCaMP6s (AAV2/9-CAG-FLEX-GCaMP6s-WPRE-SV40, 1.3×10^{13} mol/ml; U. Penn) ChR2 (AAV2/9-EF1-DIO-hChR2-(H134R)-mCherry-hGH, 5.8×10^{13} ; U. Penn), ArchT (AAV2/9-CAG-FLEX-ArchT-tdTom, 1.5×10^{13} ; U. North Carolina), eNpHR3.0 (AAV2/1-EF1-DIO-eNpHR3.0-EYFP, 2.8×10^{13} ; U. Penn) or hM3Dq (AAV2-hSyn-DIO-hM3D(Gq)-

mCherry, 4×10^{12} ; U. North Carolina) were injected bilaterally into the mouse ARN. Animals were given Carprofen (5mg/kg b.w. s.c), anesthetized with 2% isoflurane, and placed in a stereotaxic apparatus. A custom-made bilateral Hamilton syringe apparatus holding two 1 μ l Hamilton syringes with 25-gauge needles held 0.9 mm apart was used to perform bilateral injections. Each syringe was filled with 1.0 μ l of AAV. For the GCaMP6 and ChR2 experiments, mice received a single bilateral injection of AAV 1.2mm posterior to bregma, and 6.0mm deep. The needles were lowered into place and after 3 min, 1 μ l of AAV injected over 10 min, and the needles left *in situ* for a further 10 min before being withdrawn. For the ArchT and eNpHR3.0 experiments mice received two bilateral injections (1 μ l per injection per hemisphere) at 1.0 and 1.2mm posterior to Bregma, and 5.7mm depth. Some mice received bilateral 1 μ l co-injections of GCaMP6s and hM3Dq AAVs.

Brain slice electrophysiology and calcium imaging:

Electrophysiological recordings of ARN^{KISS} neurons were carried out as described previously (27). Briefly, coronal brain slices (250 μ m thick) were placed under an upright epifluorescence microscope and constantly perfused (1.5mL/min) with 95%O₂/5%CO₂ equilibrated 30°C artificial cerebrospinal fluid (aCSF; in mM): NaCl 120, KCl 3, NaHCO₃ 26, NaH₂PO₄ 1, CaCl₂ 2.5, MgCl₂ 1.2 and glucose 10. Neurons expressing GFP or tdTomato were first visualized by brief fluorescence illumination, and subsequently approached using infrared differential interference contrast optics. ARN^{KISS} neuron action potential firing was recorded in voltage clamp mode using the cell-attached loose patch configuration. Recording electrodes (3-5 M Ω) were filled with aCSF including 10 mM HEPES and low resistance seals (10-30 M Ω) achieved. Electrophysiological signals were recorded using a Multiclamp 700B amplifier connected to a Digidata 1440A digitizer (Molecular Devices). Signals were low-pass filtered at 3 kHz before being digitized at a rate of 10 kHz. Signal acquisition was carried out with pClamp 10 (Molecular Devices). Spontaneous spikes were detected using the threshold-crossing method in pClamp 10. Action potential firing was monitored for >2 min before 5-min illumination with a 532nm laser (0.6mW, Ikecool) delivered through a fiber optic (200 μ m diameter, NA 0.37; Doric Lenses) placed above the surface of the slice, near the recorded cell. For chemogenetic experiments, action potential firing of ARN^{KISS} neurons was monitored for >3 minutes before CNO (1-10 μ M) was bath-applied for 10 minutes. Statistical comparisons were undertaken using the non-parametric Friedman tests with the sample size used in tests being the number of neurons.

Brain slice calcium imaging was carried out as previously reported (42). In brief, brain slices (as above) were illuminated through a 40X immersion objective, using the xenon arc light source (300 W; filtered by a GFP filter cube, excitation 460-480 nm, Olympus) and the shutter of a Lambda DG-4 (Sutter Instruments). Epifluorescence (485 nm long pass and emission 495-540 nm) was collected using a Hamamatsu ORCA-ER digital CCD camera. A focal plane including several fluorescent ARN^{KISS} somata was chosen and acquisitions (100 ms light exposure at 2 Hz for 10-20 min) started. For analysis, time-series of images were processed in ImageJ. Regions of interest comprised of individual in-focus fluorescent ARN^{KISS} somata with average fluorescence intensity measured in each frame. Fluorescence

intensity data were analyzed using scripts written in R (<http://www.r-project.org/>) or using pClamp 10. Relative fluorescence changes were calculated using the following formula: $\Delta F/F = (F_t - F)/F * 100$ where F is the baseline fluorescence intensity calculated as the mean fluorescence intensity over a 2-min period preceding drug applications and F_t is the test fluorescence.

Fiber photometry:

Following AAV injection of GCaMP6s, an optical fiber (400 μ m diameter, 0.48 NA; Doric Lenses, Canada) was positioned immediately above the mid-caudal ARN and the mouse GDX. Mice were housed individually and allowed to recover for at least three weeks with daily handling and habituation to the fiber photometry recording set-up. Over a period of 3-10 weeks following surgery, fluorescence signals were recorded from freely-behaving mice in their home cages. Mice injected with GCaMP6s and hM3Dq AAVs received a single injection of CNO (2mg/kg, s.c.) during the recording.

Fluorescence signals were acquired using a fiber photometry system based on a previous published design (20). All optical components were purchased from Doric Lenses. Excitation light was provided by violet (referred to as 405nm) and blue (referred to as 490nm) fiber-coupled LEDs which were sinusoidally modulated at 531 and 211Hz respectively. Each excitation line passed through an excitation filter (400-410nm and 460-490nm) and focused into a 400 μ m optic fiber which connected to the mouse. Emitted fluorescence was collected by the same fiber, passed through a 500-550nm emission filter and focused onto a photoreceiver (2151 Newport). Two GCaMP6s emission signals were recovered by demodulating the 405nm (531Hz) and 490nm (211Hz) signals. The 405nm signal reported the excitation/emission of GCaMP6s at its isosbestic point and was used as a background (non-calcium-dependent) signal. The 490nm signal, reported calcium-dependent GCaMP6s excitation/emission and was used to infer changes in $i[Ca^{2+}]$. The signals were modulated/demodulated and recorded at 10Hz with custom software (Tussock Innovation, Otago, NZ). Signals from the 405nm control channel were subtracted from the 490 channel before analysis. Photometry data could either be collected continuously or in 5s on/15s off scheduled-mode. The power output at the tip of the fiber was set at 50 μ W. Fluorescence signals (490-405) were collected and converted to $\Delta F/F$ (%) values as follows: $\Delta F/F = 100 * (F - F_b)/F$ where F_b was the basal fluorescence signal between events and F the recorded fluorescence.

To assess the relationship of calcium events with pulsatile LH secretion, freely-behaving mice were attached to the fiber photometry system and LH secretion measured by obtaining 3 μ L blood samples every 3-6 min from the tail-tip over a period of 60-80 min. Blood sampling and subsequent ELISA measurement of LH was undertaken as reported previously(12, 23). The assay sensitivity was 0.04ng/mL, with intra- and inter- assay coefficients of variation of 9.3% and 10.5%, respectively.

Optogenetic experiments:

Adult control and experimental mice were implanted with indwelling unilateral

or bilateral optic fibres (200 μ m diameter, pitch 0.9mm, 0.37 NA; Doric Lenses) at the time of AAV injection. Fibres were implanted 1.1mm posterior to Bregma, centered around the sagittal sinus, 5.7mm deep. Mice were then habituated for 2-5 weeks prior to undergoing the optogenetic activation and bleeding protocol.

For ChR2 optogenetic studies, intact diestrous female and male mice were connected to the laser and 15-min later given 5-ms pulses of blue light (473nm, 5mW; DPSS laser, Ikecool, USA) for one min at a frequency of 10 Hz, repeated three times over 30-min intervals. Serial 3 μ L blood samples were collected from the tail tip at -12, -6, 0, 3, 6, 9, 12, 18, 24 and 30 min, where 0 was the start of the laser stimulation. Statistical analysis of evoked LH values was undertaken using one-way repeated-measures ANOVA with Dunnett's *post hoc* test, comparing values to the -6min LH level.

For inhibitory optogenetic experiments, mice GDX at the time of AAV injection were attached to the laser patch cord in their home cages, left to acclimate for 15 min, and 5-min blood sampling (3 μ L per sample) undertaken for 180 min as described above. The laser was turned on for a 30-min period after 90 min of baseline sampling. Illumination with constant green light from a DPSS laser (532nm, ~5mW per fibre; Ikecool) was controlled by a Master9 stimulator. As a control in ArchT experiments, 4 mice underwent a second bleeding experiment, at least 2 weeks after the first, in which they received blue light illumination (472nm). Analysis of the frequency and amplitude of pulses was determined for the 0-85min baseline period, for the 30min laser illumination period, and for the two 30min periods thereafter. Taking insight from the fiber photometry results, an LH pulse in GDX mice was defined as a peak level of LH >10% above the preceding value. For each detected pulse, the amplitude was determined by subtracting the highest LH value from the basal value immediately prior to the onset of the pulse. Data were compared using one-way repeated measures ANOVA with *post-hoc* Dunnett's tests comparing the baseline with and post-illumination time periods.

Immunohistochemistry:

Immunofluorescence was undertaken as reported previously (24). Coronal brain sections (30 μ m thick) covering the full length of the ARN were used to determine the locations of optic fibers and/or processed for dual-label analysis. Antisera against GFP (1:5000; Aves Labs Inc.) was used to detect GCaMP6. Dual fluorescence images were captured on a NikonA1+ inverted confocal microscope. Two sections at each of the rostral, middle and caudal levels of the ARN (Suppl. Fig.5) were analyzed in each mouse by counting the total number of cells that expressed τ GFP or tdTomato, reporting kisspeptin expression, and either GFP (GCaMP6), tdTomato (ArchT), or eYFP (eNpHR3.0).

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References

1. Le Tissier P, Campos P, Lafont C, Romano N, Hodson DJ, & Mollard P (2017) An updated view of hypothalamic-vascular-pituitary unit function and plasticity. *Nature reviews. Endocrinology* 13(5):257-267.
2. Lightman SL & Conway-Campbell BL (2010) The crucial role of pulsatile activity of the HPA axis for continuous dynamic equilibration. *Nat Rev Neurosci* 11(10):710-718.
3. Leng G & Brown D (1997) The origins and significance of pulsatility in hormone secretion from the pituitary. *J Neuroendocrinol* 9(7):493-513.
4. Russell GM, Kalafatakis K, & Lightman SL (2015) The importance of biological oscillators for hypothalamic-pituitary-adrenal activity and tissue glucocorticoid response: coordinating stress and neurobehavioural adaptation. *J Neuroendocrinol* 27(6):378-388.
5. Dierschke DJ, Bhattacharya AN, Atkinson LE, & Knobil E (1970) Circoral oscillations of plasma LH levels in the ovariectomized rhesus monkey. *Endocrinology* 87(5):850-853.
6. Belchetz PE, Plant TM, Nakai Y, Keogh EJ, & Knobil E (1978) Hypophysial responses to continuous and intermittent delivery of hypophyseal gonadotropin-releasing hormone. *Science* 202(4368):631-633.
7. Knobil E (1980) The neuroendocrine control of the menstrual cycle. *Recent Prog Horm Res* 36:53-88.
8. Goodman RL & Karsch FJ (1981) The hypothalamic pulse generator: a key determinant of reproductive cycles in sheep. *Biological Clocks in Seasonal Reproductive Cycles*, ed Follett B.K. FDE (John Wright & Sons, Ltd, Bristol, England), Vol Colson Papers No. 32, pp 223-236.
9. Seminara SB, Messenger S, Chatzidaki EE, Thresher RR, Acierno JS, Jr., Shagoury JK, Bo-Abbas Y, Kuohung W, Schwinof KM, Hendrick AG, Zahn D, Dixon J, Kaiser UB, Slaugenhaupt SA, Gusella JF, O'Rahilly S, Carlton MB, Crowley WF, Jr., Aparicio SA, & Colledge WH (2003) The GPR54 gene as a regulator of puberty. *N Engl J Med* 349(17):1614-1627.
10. de Roux N, Genin E, Carel JC, Matsuda F, Chaussain JL, & Milgrom E (2003) Hypogonadotropic hypogonadism due to loss of function of the KiSS1-derived peptide receptor GPR54. *Proc Natl Acad Sci U S A* 100(19):10972-10976.
11. Kirilov M, Clarkson J, Liu X, Roa J, Campos P, Porteous R, Schutz G, & Herbison AE (2013) Dependence of fertility on kisspeptin-Gpr54 signaling at the GnRH neuron. *Nature Commun* 4:2492.
12. Steyn FJ, Wan Y, Clarkson J, Veldhuis JD, Herbison AE, & Chen C (2013) Development of a methodology for and assessment of pulsatile luteinizing hormone secretion in juvenile and adult male mice. *Endocrinology* 154(12):4939-4945.
13. Uenoyama Y, Nakamura S, Hayakawa Y, Ikegami K, Watanabe Y, Deura C, Minabe S, Tomikawa J, Goto T, Ieda N, Inoue N, Sanbo M, Tamura C,

- Hirabayashi M, Maeda KI, & Tsukamura H (2015) Lack of pulse and surge modes and glutamatergic stimulation of luteinising hormone release in Kiss1 knockout rats. *J Neuroendocrinol* 27(3):187-197.
14. Herbison AE (2016) Control of puberty onset and fertility by gonadotropin-releasing hormone neurons. *Nature reviews. Endocrinology* 12(8):452-466.
 15. Lehman MN, Coolen LM, & Goodman RL (2010) Minireview: Kisspeptin/Neurokinin B/Dynorphin (KNDy) cells of the arcuate nucleus: A central node in the control of gonadotropin-releasing hormone secretion. *Endocrinology* 151(8):3479-3489.
 16. Navarro VM, Gottsch ML, Chavkin C, Okamura H, Clifton DK, & Steiner RA (2009) Regulation of gonadotropin-releasing hormone secretion by kisspeptin/dynorphin/neurokinin B neurons in the arcuate nucleus of the mouse. *J Neurosci* 29(38):11859-11866.
 17. de Croft S, Boehm U, & Herbison AE (2013) Neurokinin B activates arcuate kisspeptin neurons through multiple tachykinin receptors in the male mouse. *Endocrinology* 154(8):2750-2760.
 18. Qiu J, Nestor CC, Zhang C, Padilla SL, Palmiter RD, Kelly MJ, & Ronnekleiv OK (2016) High-frequency stimulation-induced peptide release synchronizes arcuate kisspeptin neurons and excites GnRH neurons. *eLife* 5:e16246.
 19. Wakabayashi Y, Nakada T, Murata K, Ohkura S, Mogi K, Navarro VM, Clifton DK, Mori Y, Tsukamura H, Maeda K, Steiner RA, & Okamura H (2010) Neurokinin B and dynorphin A in kisspeptin neurons of the arcuate nucleus participate in generation of periodic oscillation of neural activity driving pulsatile gonadotropin-releasing hormone secretion in the goat. *J Neurosci* 30(8):3124-3132.
 20. Lerner TN, Shilyansky C, Davidson TJ, Evans KE, Beier KT, Zalocusky KA, Crow AK, Malenka RC, Luo L, Tomer R, & Deisseroth K (2015) Intact-Brain Analyses Reveal Distinct Information Carried by SNc Dopamine Subcircuits. *Cell* 162(3):635-647.
 21. Yeo SH, Kyle V, Morris PG, Jackman S, Sinnett-Smith LC, Schacker M, Chen C, & Colledge WH (2016) Visualisation of Kiss1 neurone distribution using a Kiss1-CRE transgenic mouse. *J Neuroendocrinol* 28(11).
 22. Alexander GM, Rogan SC, Abbas AI, Armbruster BN, Pei Y, Allen JA, Nonneman RJ, Hartmann J, Moy SS, Nicolelis MA, McNamara JO, & Roth BL (2009) Remote control of neuronal activity in transgenic mice expressing evolved G protein-coupled receptors. *Neuron* 63(1):27-39.
 23. Czielesky K, Prescott M, Porteous R, Campos P, Clarkson J, Steyn FJ, Campbell RE, & Herbison AE (2016) Pulse and Surge Profiles of Luteinizing Hormone Secretion in the Mouse. *Endocrinology* 157(12):4794-4802.
 24. Han SY, McLennan T, Czielesky K, & Herbison AE (2015) Selective optogenetic activation of arcuate kisspeptin neurons generates pulsatile luteinizing hormone secretion. *Proc Natl Acad Sci U S A* 112(42):13109-13114.
 25. Chow BY, Han X, Dobry AS, Qian X, Chuong AS, Li M, Henninger MA, Belfort GM, Lin Y, Monahan PE, & Boyden ES (2010) High-performance

- genetically targetable optical neural silencing by light-driven proton pumps. *Nature* 463(7277):98-102.
26. Mattis J, Tye KM, Ferenczi EA, Ramakrishnan C, O'Shea DJ, Prakash R, Gunaydin LA, Hyun M, Fenno LE, Gradinaru V, Yizhar O, & Deisseroth K (2011) Principles for applying optogenetic tools derived from direct comparative analysis of microbial opsins. *Nat Methods* 9(2):159-172.
 27. de Croft S, Piet R, Mayer C, Mai O, Boehm U, & Herbison AE (2012) Spontaneous kisspeptin neuron firing in the adult mouse reveals marked sex and brain region differences but no support for a direct role in negative feedback. *Endocrinology* 153(11):5384-5393.
 28. Qiu J, Nestor CC, Zhang C, Padilla SL, Palmiter RD, Kelly MJ, & Ronnekleiv OK (2016) High-frequency stimulation-induced peptide release synchronizes arcuate kisspeptin neurons and excites GnRH neurons. *eLife* 5.
 29. Herbison AE (2015) Physiology of the adult GnRH neuronal network. *Knobil and Neill's Physiology of Reproduction*, eds Plant TM & Zeleznik AJ (Academic Press), 4th Ed Vol 1, pp 399-467.
 30. Sternson SM & Eiselt AK (2017) Three Pillars for the Neural Control of Appetite. *Annu Rev Physiol* 79:401-423.
 31. Fu LY & van den Pol AN (2010) Kisspeptin directly excites anorexigenic proopiomelanocortin neurons but inhibits orexigenic neuropeptide Y cells by an indirect synaptic mechanism. *J Neurosci* 30(30):10205-10219.
 32. Hu MH, Li XF, McCausland B, Li SY, Gresham R, Kinsey-Jones JS, Gardiner JV, Sam AH, Bloom SR, Poston L, Lightman SL, Murphy KG, & O'Byrne KT (2015) Relative Importance of the Arcuate and Anteroventral Periventricular Kisspeptin Neurons in Control of Puberty and Reproductive Function in Female Rats. *Endocrinology* 156(7):2619-2631.
 33. Lehman MN, Hileman SM, & Goodman RL (2013) Neuroanatomy of the kisspeptin signaling system in mammals: comparative and developmental aspects. *Adv Exp Med Biol* 784:27-62.
 34. Raimondo JV, Kay L, Ellender TJ, & Akerman CJ (2012) Optogenetic silencing strategies differ in their effects on inhibitory synaptic transmission. *Nat Neurosci* 15(8):1102-1104.
 35. Yip SH, Boehm U, Herbison AE, & Campbell RE (2015) Conditional viral tract tracing delineates the projections of the distinct kisspeptin neuron populations to GnRH neurons in the mouse. *Endocrinology* 156(7):2582-2594.
 36. Herde MK, Iremonger KJ, Constantin S, & Herbison AE (2013) GnRH neurons elaborate a long-range projection with shared axonal and dendritic functions. *J Neurosci* 33(31):12689-12697.
 37. Iremonger KJ, Porteous R, & Herbison AE (2017) Spike and neuropeptide-dependent mechanisms control GnRH neuron nerve terminal Ca²⁺ over diverse time scales. *J Neurosci* 37(12):3342-3351.
 38. Glanowska KM & Moenter SM (2015) Differential regulation of GnRH secretion in the preoptic area (POA) and the median eminence (ME) in male mice. *Endocrinology* 156(1):231-241.
 39. Ramaswamy S, Guerriero KA, Gibbs RB, & Plant TM (2008) Structural interactions between kisspeptin and GnRH neurons in the mediobasal hypothalamus of the male rhesus monkey (*Macaca mulatta*) as revealed

- by double immunofluorescence and confocal microscopy. *Endocrinology* 149(9):4387-4395.
40. Keen KL, Wegner FH, Bloom SR, Ghattei MA, & Terasawa E (2008) An increase in kisspeptin-54 release occurs with the pubertal increase in luteinizing hormone-releasing hormone-1 release in the stalk-median eminence of female rhesus monkeys in vivo. *Endocrinology* 149(8):4151-4157.
 41. Ainsworth M, Lee S, Cunningham MO, Traub RD, Kopell NJ, & Whittington MA (2012) Rates and rhythms: a synergistic view of frequency and temporal coding in neuronal networks. *Neuron* 75(4):572-583.
 42. Piet R, Fraissenon A, Boehm U, & Herbison AE (2015) Estrogen permits vasopressin signaling in preoptic kisspeptin neurons in the female mouse. *J Neurosci* 35(17):6881-6892.

Figure Legends

Figure 1. ARN^{KISS} neurons exhibit periodic synchronizations in intracellular calcium concentrations tightly correlated with pulsatile LH secretion in gonadectomized male mice.

A. KISS1-Cre mice injected with Cre-dependent GCaMP6s AAV into the ARN (green) and connected to a fiber photometry system with regular blood sampling from the tail tip. Alternating 490 nm (Ca^{2+} dependent) and 405-nm (Ca^{2+} independent) wavelength light passes down the photometry fiber. **B.** Simultaneous GCaMP6s calcium (top) and cell-attached (bottom) recordings from an ARN^{KISS} neuron. numbers = spikes per burst. **C.** Fluorescence emission traces (490 nm calcium=blue, 405 nm background = purple) recorded from ARN^{KISS} neurons of a gonadectomized male mouse with the subtracted signal (490 – 405) shown below. **D.** Continuous recording of two calcium events from a gonadectomized mouse. **E.** Average fluorescence waveform of calcium events (16 events, 7 mice) normalized to beginning of each event. **F.** Simultaneous brain slice recording of GCaMP6s fluorescence (top) and action potential firing (bottom) from a GCaMP6s/hM3Dq-expressing ARN^{KISS} neuron showing the response to CNO applied in the bath. **G.** Representative trace showing effect of i.p. CNO (arrow) on GCaMP6 calcium signal from ARN^{KISS} neurons in a gonadectomized mouse. **H & I.** Traces showing the near perfect correlation of ARN^{KISS} neuron GCaMP6 calcium events (black) with pulses of LH secretion (red) in a gonadectomized male mouse. **I.** Shows the mouse with the slowest calcium events.

Figure 2. Brief 1-min channelrhodopsin activation of ARN^{KISS} neurons evokes pulsatile LH secretion in freely behaving intact male and female mice.

A. Schematic of experiment. **B.** Repeated 1-min blue light (473nm) activations of ARN^{KISS} neurons evoke LH pulses in an intact male mouse. **C.** Mean ($\pm$ SEM) LH levels (red symbols indicate significantly elevated LH levels; repeated measures ANOVA with Dunnett's *post-hoc* tests; n=5). **D.** Repeated 1-min blue light (473 nm) activations of ARN^{KISS} neurons evoke LH pulses in an intact female mouse. **E.** Mean ($\pm$ SEM) LH levels (red symbols indicate significantly elevated LH levels; repeated measures ANOVA with Dunnett's *post-hoc* tests; n=6). **F.** Overlaid profiles of endogenous (red) and ChR2-activated (blue) LH pulses in intact male mice. Pulses normalized for fold-increase over baseline; endogenous LH pulses (n=6) sampled every 10 min and ChR2-activated LH pulses (n=15) every 6 min.

Figure 3. ArchT inhibition of ARN^{KISS} neurons suppresses pulsatile LH secretion in gonadectomized female mice.

A. KISS1-Cre mice injected with Cre-dependent ArchT-tdTomato AAV into the ARN (red) and implanted with bilateral optic fibers in the ARN. **B.** Dual fluorescence image of ARN^{KISS} neurons expressing ArchT (red) and kisspeptin (GFP reporter). **C.** Action potential firing in an ArchT-expressing ARN^{KISS} neuron in the presence of 10nM NKB in a brain slice. The neuron responds to green light

illumination (shading) with a decrease in firing followed by a return to control levels. Top trace, action potential firing. Bottom trace, rate meter for the same cell. **D.** Mean ($\pm$ SEM) normalized firing rate of ARN^{KISS} neurons (n=11) responding to green light. * $p < 0.05$ compared with baseline (Friedman test). **E-H,** Pulsatile LH secretion in ArchT KISS1-Cre (E-G) and wild-type (H) gonadectomized female mice. Green light illumination (30 min) indicated by green shading, LH pulses by asterisks. Trace in **G.** shows the same mouse in **F.** but illuminated with blue light. **I.** Mean ($\pm$ SEM) LH levels in KISS1-Cre mice (n=6) showing the suppression of LH secretion during green light illumination and subsequent slow recovery. **J & K,** Mean ($\pm$ SEM) LH pulse amplitude and pulse frequency prior to (0-85 min), during (90-115 min; green shading) and subsequent (120-145 and 150-175 min) to green laser illumination. * $p < 0.05$, ** $p < 0.01$ versus 0-85min; ANOVA with Dunnett's *post hoc* tests, n=6. **L-N,** Basal LH levels, LH pulse amplitude and frequency in control mice including blue light-illuminated KISS1-Cre and wild-type AAV-injected mice (n=8).

Figure 4. Halorhodopsin inhibition and activation of ARN^{KISS} neurons suppresses and then initiates on-going pulsatile LH secretion in gonadectomized female mice.

A. KISS1-Cre mice injected with Cre-dependent eNpHR3.0-eYFP AAV into the ARN (green) and implanted with bilateral optic fibers in the ARN. **B.** Dual fluorescence image of ARN^{KISS} neurons expressing eNpHR3.0 (green) and kisspeptin (tdTomato reporter). **C.** Action potential firing in an eNpHR3.0-expressing ARN^{KISS} neuron in the presence of 20nM NKB. The neuron responds to green light with an abrupt decrease in firing followed by a sharp rebound activation immediately after the light is stopped. Top trace, action potential firing. Bottom trace, rate meter. **D.** Mean ($\pm$ SEM) normalized firing rate of ARN^{KISS} neurons (n=6) responding to green light. * $p < 0.05$ compared with baseline (Friedman test). **E-H,** Pulsatile LH secretion in eNpHR3.0 KISS1-Cre (E,F) and wild-type (G,H) gonadectomized female mice. Green light illumination indicated by green shading and LH pulses by asterisks. **I.** Mean ($\pm$ SEM) LH levels in KISS1-Cre mice (n=8) showing the suppression of LH secretion during green light illumination and subsequent rebound of LH and re-setting of pulse generator to evoke a subsequent LH pulse in all mice (arrow). **J & K,** Mean ($\pm$ SEM) LH pulse amplitude and pulse frequency prior to (0-85 min), during (90-115 min; green shading) and subsequent (120-145 and 150-175 min) to green laser illumination. * $p < 0.05$, ** $p < 0.01$ versus 0-85min; ANOVA with *post-hoc* Dunnett's test, n=8. **L-N,** Basal LH levels, LH pulse amplitude and frequency in control AAV-injected wild-type mice (n=7).